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FUSION SYSTEM HAVING TARGET TRACKING

Abstract

A fusion system and method generates an energy beam based on a trajectory of a target. The fusion system includes a reaction chamber, an energy pulse source, and a tracking system. The tracking system generates tracking data based on the trajectory of the target outside of the reaction chamber, and the energy pulse source is activated in response to determining that the trajectory is directed to a strike zone inside of the reaction chamber. The energy pulse source generates the energy beam to strike the target.

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Background/Summary

[0001] This application claims the benefit of priority from U.S. Provisional Application No. 63/559,823, titled “Fusion System Having Target Tracking” and filed Feb. 29, 2024, which is incorporated herein by reference in its entirety.

FIELD

[0002] The present disclosure relates to fusion systems. More particularly, the present disclosure relates to fusion systems that direct an energy beam toward a target to generate energy.

BACKGROUND

[0003] Fusion systems can be used for research purposes in high-energy physics, and are being investigated for use in energy production. Some fusion systems direct a higher-power source of energy, such as a laser beam, to a target containing fusion fuel. The target is typically positioned or launched into a containment chamber at a low pressure relative to ambient. The laser beam strikes the target to compress, heat, and ignite the target. When the target is sufficiently imploded in this way a fusion process begins. The fusion process can generate energy that may be harnessed, e.g., converted into electricity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 depicts a schematic view of a fusion system, in accordance with an embodiment.

[0005] FIG. 2 depicts a side view of a beam steering system, in accordance with an embodiment.

[0006] FIG. 3 depicts a cross-sectional view of a target injection into a chamber, in accordance with an embodiment.

[0007] FIG. 4 depicts a cross-sectional view of a shutter apparatus in an open state, in accordance with an embodiment.

[0008] FIG. 5 depicts a cross-sectional view of a shutter apparatus in a closed state, in accordance with an embodiment.

[0009] FIG. 6 depicts a cross-sectional view of a target injection into a chamber, in accordance with an embodiment.

[0010] FIG. 7 is a flowchart of a method of generating an energy beam based on target tracking, in accordance with an embodiment.

[0011] FIG. 8 is a flowchart of a method of steering an energy beam based on target tracking, in accordance with an embodiment.

[0012] FIG. 9 is a block diagram of a computing device that may perform one or more method operations, in accordance with an embodiment.

[0013] FIG. 10 depicts an energy beam, including a cross-sectional view of the energy beam, in accordance with an embodiment.

DETAILED DESCRIPTION

[0014] The following description sets forth numerous specific details such as examples of specific systems, components, methods, and so forth, in order to provide a good understanding of embodiments of the present invention. It will be apparent to one skilled in the art, however, that at least some embodiments of the present invention may be practiced without these specific details. In other instances, well-known components or methods are not described in detail in order to avoid unnecessarily obscuring the description of the exemplary embodiments. Thus, the specific details set forth are merely exemplary. Particular implementations may vary from these exemplary details and still be contemplated to be within the spirit and scope of the present invention.

[0015] To be commercially viable, laser-driven fusion systems are expected to operate at a sufficient repetition rate to generate energy goals. Accordingly, an energy beam, e.g., a laser beam, can precisely and repeatably, in a deterministic manner, engage targets in a reaction chamber to

efficiently generate energy. Achieving such precision focused laser-target engagement has several fundamental technical challenges, as described below.

[0016] A first technical challenge is tracking and precisely engaging the target with focused laser pulses at a predetermined location within a reaction chamber. For example, the target can include a moving hydrogen (deuterium-tritium) (DT) target pellet that focused laser pulses are to engage at a center of the reaction chamber. The DT pellet may be shot toward the center from a mechanism located external to the reaction chamber. To achieve fusion, a series of focused laser pulses may be required to hit the center of the moving DT pellet with an accuracy of 25 microns. Achieving such accuracy is challenging and requires tracking of the target.

[0017] A second technical challenge is compensating for wavefront error (WFE) in the laser pulses to ensure diffraction-limited focus profiles engage the DT pellet at the center of the reaction chamber.

[0018] A third technical challenge is ensuring that a final amplified beam does not encounter anything physical except the target at the center of the reaction chamber. To effectively engage the target, laser beams may be amplified to the final amplified beam beyond a fluence threshold. WFE compensation mechanisms, e.g., optical elements or coatings, cannot operate effectively about the fluence threshold, e.g., the final amplified beam can damage the mechanisms.

[0019] A fourth technical challenge is maintaining vacuum of the reaction chamber when passing the laser pulses toward the target. Energies of the focused engagement laser pulses, which are required to initiate fusion, can break down a gas-filled laser transport medium. Accordingly, the final laser propagation distance towards focus, after final amplification, must occur in vacuum. A direct line of sight into the vacuum may also be needed to facilitate precision beam steering and WFE compensation prior to arrival at target in the chamber center. Achieving line of sight passage of laser pulses into a vacuum without intermediate optical elements is a challenge.

[0020] A fifth technical challenge is time synchronization of the firing of front-end laser sources with beam pointing and WFE compensation. Such timing can be critical to the laser pulses arriving at the target when planned to initiate fusion. Achieving synchronization while the target is in motion to the chamber center is challenging.

[0021] A sixth technical challenge is providing fail-safes to ensure that the amplified high energy laser pulses do not engage the target if precision targeting cannot be achieved. More particularly, avoiding laser pulse generation when fusion cannot be effectively initiated can provide a more efficient process. Achieving real-time fail-safes to avoid misfires is challenging.

[0022] A commercially viable fusion system is described below, which can overcome the above-listed technical challenges. The fusion system can provide a precise launch of each target in addition to active closed-loop tracking and beam steering capability to compensate for any variability in trajectory between successive shots. The system can incorporate a shutter apparatus to provide a sufficient line of sight time into the reaction chamber to the moving target as the target approaches the chamber center. The system can provide sufficient temporal bandwidth for fail-safe operation to avoid unnecessary laser pulse generation and to steer the laser beam in a time-synchronized manner to engage the target at the chamber center. The system can provide WFE compensation (in addition to beam steering). The system may lack optical elements or physical structures that impede or contact the final amplified beam after the last amplification stage to reduce a likelihood of damage to optical elements and coatings.

[0023] Referring to FIG. 1, a schematic view of a fusion system is shown in accordance with an embodiment. A fusion system **100** can include an energy pulse source **102** to generate an energy beam **104** that can be directed to a target **106**. The fusion system **100** can be, for example, an inertial confinement fusion (ICF) system or an inertial fusion energy (IFE) system. The ICF system may operate on a single-shot, non-repetitive basis. Alternatively, the IFE system may operate continuously, in a pulsed fashion, e.g., at a frequency of one shot per second. Accordingly, the fusion system may have an architecture to perform a fusion process.

[0024] The fusion system **100** may be more broadly described as a target tracking system **108**, in which the energy pulse source **102** includes a laser pulse source **110** to generate the energy beam **104**, e.g., a laser beam **112**. The laser beam **112** can be directed to the target **106**. It will be appreciated then, that although the target tracking system **108** may be used in a fusion process and may thus be termed the fusion system **100**, the target tracking system **108** may be used in other applications that require precision focused-laser target engagement.

[0025] The energy pulse source **102** can include one or more front end laser sources to generate a grid of co-aligned beamlets (FIG. **10**). The beamlets can be square beams having rounded edges that combined to form the energy beam **104**. In an embodiment, the front end laser sources are excimer lasers. Accordingly, the energy beam **104** can include light at wavelengths of 248 nm or 193 nm.

[0026] The beamlets can pass through amplification operations and additional beamlines can be combined to output, from the energy pulse source **102**, the energy beam **104** as an $m \times n$ grid of co-aligned, square, time-staggered collimated beamlets having respective side lengths. For example, the energy beam **104** can include a collimated 3×2 beam array having beamlets with cross-sections measuring 25 cm by 25 cm. The energy beam **104** having the beamlets can enter an optical path **114** of the fusion system **100**, the path being directed toward a reaction chamber **116** of the system. The reaction chamber **116** can be a vacuum chamber **118** within which a fusion reaction occurs, for example.

[0027] The collimated beamlets of the energy beam **104** can enter the optical path **114** after pre-amplification. When entering the optical path **114**, the beamlets can be timed. For example, the beamlets may be time-gapped by a predetermined lag. In an embodiment, a delay of 10-15 ns, e.g., 12 ns, exists between the entry of each beamlet into the optical path **114**.

[0028] The beamlets can propagate along the optical path **114** and pass through a phase plate array **120**. The phase plate array **120** can include a grid of $m \times n$ phase plates, matching the grid of beamlets, to impart an individual phase prescription to each beamlet. Each beamlet of the energy beam **104** can pass through a respective phase plate of the phase plate array **120** to develop an optimized intensity profile pattern during final focus of the beamlet. More particularly, phase plate array **120** can put a prescribed phase on each beamlet to control an intensity profile or intensity distribution of the energy beam **104** at the point of focus. The point of focus can be at the target **106** in the reaction chamber **116**, e.g., where the energy beam **104** strikes the target **106**. The term "strike," as used herein, can refer to the process of laser interaction with, irradiation of, or incidence on the target. The struck target can absorb the laser in a manner sufficient to achieve implosion and fusion ignition, as required by a specific target design.

[0029] After undergoing preamplification, the beamlets can transmit through a beamsplitter **121**. The beamlets can remain at a nominally uniform intensity as they pass through the beamsplitter **121**. The beamsplitter **121** can be a dichroic beamsplitter that directs a portion of light from the beamlets toward a wavefront sensor array **122**. As described below, the wavefront sensor array **122** can be a component of a beam steering system **124** used to steer the energy beam **104** toward the moving target **106**. More particularly, the beam steering system **124** can include the wavefront sensor array **122** and a deformable mirror array **126**. The wavefront sensor array **122** can detect tilt and wavefront error in the beamlets, and the deformable mirror array **126** can impart tilt and wavefront correction.

[0030] A portion of the energy beam **104** can transmit through the beamsplitter **121** toward the deformable mirror array **126**. The deformable mirror array **126** can have an $m \times n$ grid of deformable mirrors (e.g., a 3×2 grid) and, thus, each beamlet can reflect off of a respective deformable mirror to continue along the optical path **114** toward the target **106**. Each deformable mirror can include a global tip/tilt mechanism that can be moved by one or more actuators to impart phase and steering on the beamlets. Accordingly, each deformable mirror can be capable of independently imparting dynamically prescribed high order phase and global x/y tilt correction to a respective beamlet.

More particularly, the deformable mirror array **126** can correct wavefront error sensed by the wavefront sensor array **122**. Accordingly, the beam steering system **124** can meet a technical need to compensate for wavefront error in the laser pulses, prior to exceeding a threshold fluence level, to ensure diffraction limited focus profiles engage the target **106** at the center of the reaction chamber **116**.

[0031] The phase and tilt corrected beamlets can propagate forward along the optical path **114** toward a parabolic mirror **128**. The beamlets can reflect from the parabolic mirror **128** to converge toward the center of the reaction chamber **116**. The parabolic mirror **128** may be more than 25 m, e.g., 50 m, from the chamber center.

[0032] In an embodiment, the parabolic mirror **128** includes a global off-axis parabolic mirror. The parabolic mirror **128** can be a final optic in the optical path **114** between the energy pulse source **102** and the reaction chamber **116**. Accordingly, the energy beam **104** can reflect and converge with a slow focal ratio from the parabolic mirror **128** to the center of the reaction chamber **116**.

[0033] Beryllium can be selected as a material for the parabolic mirror **128**, the final focusing optic, to ensure minimal long-term damage from neutron bombardment emanating from the reaction chamber **116** at a small solid angle during fusion ignition. More particularly, only the off-axis parabolic Beryllium mirror may experience neutron bombardment, which can protect the other optics in the optical path **114** from damage.

[0034] The energy beam **104**, after reflecting forward from the off-axis parabolic mirror **128**, can have a fluence level that is less than a threshold fluence level to ensure that no laser induced damage occurs to the optical elements and coatings in the optical path **114**. The non-destructive beamlets can enter a Stimulated Brillouin scattering (SBS) laser amplifier, which can perform an amplification stage. The SBS amplifier **130** can output the energy beam **104** at a fluence level that is above the threshold fluence level. In an embodiment, the energy of each beamlet in the energy beam **104** is 1 MJ. For example, the energy beam **104** may have a 3×2 grid of beamlets, totaling an energy transfer of 6 MJ to the target for a single-sided configuration. Similarly, for a double-sided configuration (FIG. 6) having a second energy beam **104**, the total energy transfer can be 12 MJ. Accordingly, the optical path **114** may lack physical structures in the beam path between the SBS amplifier **130** and the reaction chamber **116** to avoid destruction of such structures by the energy beam **104**. After the final focusing of the off-axis parabolic mirror **128** and the SBS amplifier **130**, the beamlets can converge to an f-number of 150 to 250, e.g., 200. The energy beam **104** can therefore travel to engage target **106** at the final focus in the chamber center without encountering physical structures, including optical elements.

[0035] In an embodiment, the energy beam **104** passes forward through a shutter apparatus **132**. The shutter apparatus **132** can include a vacuum shutter. As described below with respect to FIGS. 4-5, the shutter apparatus **132** can permit passage of the energy beam **104** from a higher pressure zone to a lower pressure zone without optical elements. More particularly, the energy beam **104** can pass through free space within the shutter apparatus **132** without striking structures of the shutter apparatus **132**. By way of example, the energy beam **104** can pass through a 30 cm by 20 cm or larger aperture, without touching a physical structure, and can propagate forward toward the chamber center. The vacuum shutter may be more than 10 m, e.g., 20 m, from the chamber center.

[0036] The energy beam **104** can propagate forward from the shutter apparatus **132** toward the reaction chamber **116**. The energy beam **104** can be directed to a strike zone **134** inside of the reaction chamber **116**. For example, the energy beam **104** can pass through an entry port **136** in the reaction chamber **116**. In an embodiment, the entry port **136** includes a 25-45 cm, e.g., 32 cm, diameter aperture. The aperture can allow passage of the energy beam **104** into the reaction chamber **116**. The reaction chamber **116** may have a radius of 1-10 m, e.g., 5 m and, thus, the entry port **136** may open a small portion of the chamber wall. An energy absorbing shell **140**, formed by a flow of FLiBe (a molten salt made from a mixture of lithium fluoride (LiF) and beryllium fluoride (BeF₂)), can encompass the strike zone **134**. The fusion system **100** can include a slug

injector **137** to create the entry port **136** in the shell **140**. For example, the slug injector can include a nozzle to blast or create temporary openings in the shell **140**. The entry port **136** can extend through the energy absorbing shell **140** to allow the energy beam **104** to pass toward the strike zone **134**.

[0037] The fusion system **100** can include a target propulsion system. The target propulsion system can inject the target **106** into the reaction chamber **116**. The target propulsion system can launch the target **106**, e.g., a fusion fuel pellet, along a trajectory **142** that extends from outside the reaction chamber **116** to the strike zone **134** inside of the reaction chamber. The target injection may occur at an angle of 1-5 degrees, e.g., 1.6 degrees, from a longitudinal axis of the optical path **114** along which the energy beam **104** travels. For example, the injection location can be at a location between the entry port **136** and the shutter apparatus **132**, e.g., 10 m from the chamber center, and the target **106** can accelerate from rest at the injection location along the trajectory **142**. The target propulsion system can incorporate an air-powered propulsion system that shoots the target **106** along the trajectory **142** toward a chamber center within the strike zone **134**. Accordingly, although the trajectory **142** may have a curvilinear path along which the target **106** travels in free space, it may essentially be a straight path because the target **106** moves at high speed.

[0038] In an embodiment, the fusion system **100** includes one or more tracking systems to track motion of the target **106** along the trajectory **142**. For example, a first tracking system **144** can generate first tracking data based on the trajectory **142** of the target **106** outside of the reaction chamber **116**. As described below, a processing device of the fusion system **100** can determine, based on the first tracking data, whether the trajectory **142** is directed to the strike zone **134**. The processing device may, in response to determining that the trajectory **142** of the target **106** is directed to the strike zone **134**, activate or fire the energy pulse source **102** to generate the energy beam **104**.

[0039] The first tracking system **144** can include crossing sources or crossing cameras. More particularly, one or more crossing sources can direct light, transverse to the trajectory **142** of the target **106**, toward one or more corresponding crossing cameras. The inline crossing sources/cameras may include several crossing sources/cameras spaced at intervals. When the target **106** passes through the light, the cameras can detect the target **106**. The processing device connected to the crossing cameras can determine the trajectory or velocity of the target **106**. Accordingly, the first tracking system **144**, which can perform a first tracking stage outside of the reaction chamber **116**, can determine whether and when the target **106** will enter the strike zone **134** within the reaction chamber **116**.

[0040] The first tracking system **144** may include any tracking system that can detect the trajectory or speed of the target **106**. For example, instead of or in addition to crossing cameras, the first tracking system **144** can include a light detection and ranging (LiDAR) system. The LiDAR system can be used to track the target **106** and generate first tracking data for use in event timing.

[0041] The fusion system **100** may include a second tracking system **146**. The second tracking system **146** can include one or more light sources separate from sources of the first tracking system **144**. For example, the second tracking system **146** can include glint illumination to flood the optical path **114** with diffuse light. The glint illumination can include a bright ring illumination source between the SBS amplifier **130** and the shutter apparatus **132**. The illumination can travel to or through the entry port **136** and reflect from the moving target **106** as it moves toward a strike location **148** inside of the reaction chamber **116**. Such reflection may occur immediately after the target **106** is fired toward the chamber center, and return light **149** may propagate backward along the optical path **114**, through the shutter apparatus **132**, the SBS amplifier **130**, and the optical elements to arrive at the wavefront sensor array **122**. The wavefront sensor array **122** can detect the diffuse reflection.

[0042] In an embodiment, the second tracking system **146** generates second tracking data that, like the first tracking data, represents the trajectory **142** of the target **106**. For example, the wavefront

sensor array **122** can generate the second tracking data based on the observed return light **149**. The second tracking data, however, can represent the trajectory **142** inside of the reaction chamber **116** rather than outside of the reaction chamber **116**.

[0043] The second tracking data generated by the second tracking system **146** can be used to control beam steering of the energy beam **104**. For example, as described below, the processing device of the fusion system **100** can determine, based on the second tracking data, the strike location **148** within the strike zone **134**. The strike location **148** can be a location that the trajectory **142** of the target **106** and the optical path **114** intersects. In an embodiment, the processing device can actuate the beam steering system **124** to steer the energy beam **104** along the optical path **114** to the strike location **148**. Accordingly, the steered beam can strike the target **106** in mid-flight to initiate fusion within the strike zone **134** at the strike location **148**.

[0044] The second tracking system **146** can use back illumination when the target pellet can be adequately coated with a diffuse reflective coating to allow for direct observation by the wavefront sensor array **122**. For example, the glint illumination can include the bright ring illumination source to cause reflection from such coating, as described above. Alternatively, inline illumination sources may be placed between the mirrors of the deformable mirror array **126** to direct light along the optical path **114** to the target pellet coating. When such coating is not possible, however, alternative means of sending light backward along the optical path **114** to the wavefront sensor array **122** may be used. In an embodiment, the target pellet can self-emit light, e.g., via fluorescence or otherwise, to send light toward the wavefront sensor array **122**.

[0045] Referring to FIG. 2, a side view of a beam steering system is shown in accordance with an embodiment. In an embodiment, the second tracking system **146** includes the wavefront sensor array **122** to generate the second tracking data. The wavefront sensor array **122** can generate the second tracking data based on the return light **149** traveling proximally (backward) along the optical path **114** from the target **106**. More particularly, the return light **149** can be detected by the wavefront sensor array **122** and corresponding transduced signals can be received and processed by the processing device.

[0046] In embodiment, the beam steering system **124** includes the wavefront sensor array **122** and the deformable mirror array **126**. The deformable mirror array **126** can, as described above, include several mirrors that deform to correct errors in the energy beam **104** wavefront or steer the energy beam to impinge on the strike location **148** within the strike zone **134**. More particularly, the processing device can control actuation of the deformable mirror array **126** to direct the energy beam **104** distally (forward) to intersect the strike location **148** coincident with passage of the target **106**. The energy beam **104** can therefore initiate fusion of the target **106**.

[0047] Referring to FIG. 3, a cross-sectional view of a target injection into a chamber is shown in accordance with an embodiment. As described above, the target **106** can be injected at a target injection location **302** and propelled along the trajectory **142** through the first tracking system **144** and the entry port **136** into the reaction chamber **116**. The first tracking system **144** can detect the trajectory **142** or velocity of the target **106** outside of the reaction chamber **116**. First tracking data may be useful, for example, in determining whether the target **106** is heading toward the strike zone **134**, which may be a spatial envelope having a first size. For example, the first size may be a spherical spatial envelope having a diameter of 10-100 microns, e.g., 25 microns. Similarly, the second tracking system **146** can detect the trajectory **142** or velocity of the target **106** inside of the reaction chamber **116**. Second tracking data may be useful, for example, in determining whether the target **106** is heading toward the strike location **148**, which can be a discrete location or a spatial envelope within the strike zone **134** and having a second size smaller than the first size. For example, the second size may be a spherical spatial envelope having a diameter of 1-10 microns, e.g., 5 microns. The first tracking system **144** and the second tracking system **146** may therefore generate data useful for determining whether the trajectory **142** is directed to the strike zone **134** or strike location **148**. The beam steering system **124** can steer the energy beam **104** to the strike

location **148** based on the tracking data. In an embodiment, the trajectory **142** is detected and the energy beam **104** is steered to the strike location **148** by the beam steering system **124** while the target **106** is in flight. More particularly, determining the trajectory **142** path to the strike location **148** and steering the energy beam **104** to the strike location **148** can occur when the target **106** is traveling from the target injection location **302** outside of the reaction chamber **116** to the strike location **148** inside of the reaction chamber **116**.

[0048] The fusion system **100** can include the shutter apparatus **132** to maintain a higher pressure zone containing the energy pulse source **102** and the optical elements of the system, and a lower pressure zone containing the reaction chamber **116**. The shutter apparatus **132** can be located along the optical path **114** and may include a shutter port **402** that can cyclically or periodically actuate between an open state (FIG. 4) and a closed state (FIG. 5). When in the open state, the shutter port **402** can provide a line of sight along the optical path **114** from the final optical element of the fusion system **100** to the strike location **148** within the reaction chamber **116**. An embodiment of the shutter apparatus **132** is described below. It will be appreciated, however, that other shutter apparatus embodiments may be used to provide line of sight passage of the energy beam **104** and the return light **149** during the open state of the shutter. More particularly, the open state of the shutter apparatus **132** can be time synchronized to allow the return light **149** to travel proximally along the optical path **114** from the target **106** through the open shutter port **402** to queue the steering system, and to allow the steered energy beam **104** to travel distally along the optical path **114** through the shutter port **402** to strike the target **106** in the reaction chamber **116**. For example, suitable shutter apparatus embodiments (one of which is described below) are described in U.S. application Ser. No. 18/592,359, to Cyrus M. Herring, et al., filed on Feb. 29, 2024, entitled "Shutter Apparatus Having Ports To Control Energy Beam And Gas Transfer Between Zones," the contents of which are incorporated herein by reference. The suitable shutter apparatuses can provide line of sight passage of the energy beam **104** and return light **149** through the shutter port **402** during a same open state of the shutter apparatus **132**.

[0049] Referring to FIG. 4, a cross-sectional view of a shutter apparatus in an open state is shown in accordance with an embodiment. The shutter apparatus **132** is shown in a fully aligned, or open, state. In an embodiment, the shutter apparatus **132** includes an outer external body **404**, an outer drum **406**, a middle drum **408**, and an inner drum **410** located concentrically inside the external body **404**. The external body **404** includes the shutter port **402** exposed to a higher pressure zone upstream of the shutter apparatus **132** (toward the energy pulse source **102**) and a second shutter port **412** exposed to a lower pressure zone downstream of the shutter apparatus **132** (toward the reaction chamber **116**). The outer drum **406** includes drum ports **414A** and **414B**, the middle drum **408** includes drum ports **416A** and **416B**, and the inner drum **410** includes drum port **418A**. The body or drums can be rotatable relative to each other. For example, the outer drum **406** can be rotatable within the external body **404**. Similarly, the middle drum **408** can be rotatable within the outer drum **406**.

[0050] Rotation of the drums can align the drum ports with each other or with the shutter ports **402**, **412** of the outer external body **404**. For example, the outer drum **406** can rotate within the outer external body **404** to align the drum ports **414A**, **414B** with the shutter port **402** and the second shutter port **412**. The ports can align along a line of sight. More particularly, the shutter port **402** and the second shutter port **412** can be longitudinally aligned and separated by a distance, d , along a line of sight axis. The line of sight axis may be the axis along which the energy beam **104** and the return light **149** travel, and the shutter ports **402**, **412** can be aligned when the line of sight axis extends through the ports (allowing the energy beam **104** to pass through the aligned ports in a distal direction and the return light **149** to pass through the aligned ports in a proximal direction). When the shutter port **402** is aligned with the drum ports, the shutter port **402** can be open to the second shutter port **412** through the drum ports.

[0051] Opening the shutter port **402** to the second shutter port **412** can require additional ports of

the shutter apparatus **132** to align. For example, the middle drum **408** may rotate to move the drum port **416A** into alignment with the drum port **418A**. When the drum ports **414A**, **416A**, **418A**, **416B**, and **414B** are aligned with each other and the shutter ports **402**, **412**, along the line of sight axis, the shutter port **402** can open to the second shutter port **412** and the energy beam **104** can travel through the shutter apparatus **132** from the higher pressure atmosphere in the upstream zone to the lower pressure atmosphere in the downstream zone. Similarly, when the drum ports are aligned with each other and the shutter ports **402**, **412**, along the line of sight axis, the shutter port **402** can open to the second shutter port **412** and the return light **149** can travel through the shutter apparatus **132** from the lower pressure atmosphere to the higher pressure atmosphere

[0052] The rotational speeds of the outer drum **406**, the middle drum **408**, and the inner drum **410** can be set such that the alignment of the drum ports occurs at a predetermined periodicity. The periodicity can be selected such that alignment of the ports, or the port opening caused by such alignment, does not exist for a sufficient time to permit gas within the higher pressure atmosphere (e.g., neon or helium) to pass along the full distance, d , between the shutter ports **402**, **412**. Gas transmission is a function of the molecular velocity of the gas, e.g., $V_{\text{sub.He/Ne}}$. Thus, if the distance, d , is greater than the molecular velocity of the gas times a time, $t_{\text{sub.0}}$, during which the ports are aligned or open to each other, then the direct pathway between the gaseous and evacuated sides will not be open sufficiently long to permit gas to travel into the evacuated environment. With this in mind, transfer of gas from the first atmosphere to the second atmosphere can be minimized by observing the governing equation: $d > (V_{\text{sub.He/Ne}}) * (t_{\text{sub.o}})$. More particularly, gas transfer can be minimized by selecting an appropriate distance between the ports **402**, **412**.

[0053] Instead of or in addition to controlling distance between the shutters to limit gas transfer between atmospheres, the gas transfer may also be minimized by controlling the shutter open time of the shutter apparatus **132**. A free path transit time, $t_{\text{sub.fp}}$, of the gas across a given distance, d , can be defined as $t_{\text{sub.fp}} = d / V_{\text{sub.HeNe}}$. When the free path transit time for the gas to travel across the distance between the shutter ports **402**, **412** is less than the shutter open time of the shutter apparatus **132**, then the gas will not travel entirely through the channel into the second atmosphere. More particularly, gas transfer between the atmospheres can be minimized or eliminated by setting the shutter open time to be less than the free path transit time.

[0054] Referring to FIG. 5, a cross-sectional view of a shutter apparatus in a closed state is shown in accordance with an embodiment. The shutter apparatus **132** is in a non-aligned, or closed, state. The drums can rotate relative to each other to move respective ports into misalignment with each other. More particularly, note that the line of sight axis along which the energy beam **104** and the return light **149** are directed does not pass through the drum ports. In the non-aligned state, the ports may be closed off to each other, and neither the energy beam **104**, the return light **149**, nor the gas in the higher pressure atmosphere can travel from the shutter port **402** to the second shutter port **412**.

[0055] Based on the above description, it will be appreciated that the energy beam **104** and the return light **149** can pass through the shutter ports **402**, **412** in opposite directions during the open state of the shutter apparatus **132** and before the gas can leak from the higher pressure atmosphere to the lower pressure atmosphere. During the open time, a fast convergence of the gas may occur toward the lower pressure atmosphere. More particularly, when the shutter ports and drum ports align to create a channel for the transfer of light and gas, the gas can rush into the open channel. The surge of the gas toward the reaction chamber **116** can cause turbulence, which may cause diffraction of the energy beam **104** or the return light **149** passing through the open channel.

[0056] In an embodiment, the fusion system **100** adjusts the beam steering system **124** to correct for distortion of the energy beam **104** or the return light **149** caused by the shutter apparatus function. The surge of gas through the shutter apparatus **132** may be repeatable and deterministic, and can be characterized through computational fluid dynamics (CFD). More particularly, turbulence and a resulting light distortion effect can be characterized. The processing device can

control the steering system to adjust for the distortion effect. For example, the deformable mirror array **126** may be actuated to correct for off-set and phase distortion of the converging beamlets passing through the rushing gas as the energy beam **104** travels forward toward the strike location **148**.

[0057] Referring to FIG. **6**, a cross-sectional view of a target injection into a chamber is shown in accordance with an embodiment. The fusion system **100** may include several energy pulse sources to direct several energy beams toward the target **106**. In an embodiment, the fusion system **100** includes a second energy pulse source (not shown) to generate or fire a second energy beam **602** toward the strike zone **134** inside of the reaction chamber **116**. The second energy pulse source can direct the second energy beam **602** toward the strike location **148**, or to a second strike location offset from the strike location **148** along the trajectory **142**. The second energy beam **602** can be directed along a second optical path that is different than the optical path **114**. For example, the second energy beam **602** may enter the reaction chamber **116** on a different side than the energy beam **104** or the target **106**. Entry of the second energy beam **602** can be facilitated by similar structures to those described above, such as a second shutter apparatus or a second entry port **604** in the reaction chamber **116**. Similarly, generating, firing, or steering of the second energy beam **602** can be facilitated by similar structures to those described above, such as a second steering system.

[0058] The energy beam **104** and the second energy beam **602** can be steered to strike the target **106** at the strike location **148**. Tracking of the target **106** may be performed by the first tracking system **144** or the second tracking system **146** (not shown), and adjustments to the steering systems can be made in response to such tracking. More particularly, the energy beams **104**, **602** may be generated or fired and steered to intercept the target **106** at the chamber center to initiate fusion.

[0059] Generating (firing) and steering of the second energy pulse source and second energy beam **602** may be made based on tracking data from the first tracking system **144** or the second tracking system **146**. For example, the processing device may be configured to activate or fire the second energy pulse source when the trajectory **142** of the target **106** is directed to the strike zone **134**. Steering of the second energy beam **602** may be in response to tracking data generated by the second tracking system **146**. Alternatively, a third tracking system may be provided in line with the second optical path to generate third tracking data. The third tracking data can determine movement of the target **106** within the reaction chamber **116** in a line of sight of the second optical path. Adjustments to the second steering system can be made to steer the second energy beam **602** to intersect the trajectory **142** of the target **106** at the strike location **148**, or at the second strike location **148** offset from the strike location along the trajectory **142**.

[0060] Having described the structure of the fusion system **100** (or the target tracking system **108**), the description now turns to methods of target tracking that can be performed by the system. More particularly, energy beam control can be performed in response to tracking of the target **106** outside and inside of the vacuum chamber **118**. Such control can include fail-safes to activate or fire the energy pulse source **102** only when the target **106** is on a trajectory **142** that can be intercepted by the energy beam **104**, or steering the energy beam **104** to strike the target **106** inside of the reaction chamber **116**. Method operations, which can be performed by the fusion system components under control of the processing device, are described below.

[0061] Referring to FIG. **7**, a flowchart of a method of generating an energy beam based on target tracking is shown in accordance with an embodiment. At operation **702**, the first tracking system **144** generates first tracking data based on the trajectory **142** of the target **106** outside of the reaction chamber **116**. The first tracking data can represent an instantaneous position, velocity, or acceleration of the target **106** to allow the trajectory **142** of the target **106** and an expected future position, e.g., within the strike zone **134**, to be determined.

[0062] At operation **704**, the processing device determines, based on the first tracking data, whether the trajectory **142** is directed to the strike zone **134** inside of the reaction chamber **116**. The strike zone **134** can be the spatial envelope within which the energy beam **104** is capable of being steered

by the beam steering system **124**.

[0063] At operation **706**, the energy pulse source **102** is activated or fired to generate the energy beam **104**. Generating or firing of the energy pulse source **102** can be in response to determining that the trajectory **142** of the target **106** is directed to the strike zone **134**. More particularly, the energy pulse source **102** may not be activated or fired when the processing device determines that the trajectory **142** is not directed to the strike zone **134** inside of the reaction chamber **116**.

Accordingly, the first tracking data can be used by the processing device to make a go-no-go decision regarding whether to attempt to initiate fusion of the moving target **106**.

[0064] The determination of the trajectory **142** based on the first tracking data can act as a fail-safe to ensure general safety practices and to prevent equipment damage. When the energy beam **104** cannot intercept the target **106**, e.g., when the first tracking system **144** detects anomalies with an initial target velocity or trajectory, generating or firing of the energy pulse source **102** or the SBS amplifier **130** can be deactivated to avoid damage or increase the lifetime of the energy pulse source **102** or the reaction chamber **116**.

[0065] The fail-safe operation enabled by the first tracking data may represent a first stage of fail-safes provided by the fusion system **100**. In an embodiment, additional feedback from the wavefront sensor array **122** may facilitate a second stage of fail-safes. For example, when the wavefront sensor array detects deviations in target trajectory **142** such that achievable beam steering via the deformable mirror array **126** cannot direct the energy beam **104** to engage the target **106**, the energy pulse source **102** or the SBS amplifier **130** may be deactivated. Similarly, when the wavefront sensor array **122** detects unachievable phase compensation is required by the deformable mirror array **126** for appropriate fusion initiation, the energy pulse source **102** or the SBS amplifier **130** may be deactivated. Deactivation of the energy pulse source **102**, to not generate or fire the energy beam **104**, can be based on alternative system data, such as data generated in response to a detection of a vacuum shutter malfunction. Accordingly, the determination to activate or not activate (fire or not fire) the energy pulse source **102** may be based on the first tracking data, the second tracking data, or additional system data indicating that fusion cannot be initiated.

[0066] Referring to FIG. **8**, a flowchart of a method of steering an energy beam based on target tracking is shown in accordance with an embodiment. At operation **802**, optionally, the processing device may be configured to actuate the beam steering system **124** to a preset configuration based on the first tracking data. The first tracking system **144** can provide a rough estimation of a future position of the target **106** based on the instantaneous trajectory data. The first tracking data can be based on non-line of sight tracking and, thus, may not reliably predict the strike location **148**, however, the first tracking data may reliably predict a region within the strike zone **134** at which the target **106** will arrive. In an embodiment, the processing device can actuate the beam steering system **124** based on the first tracking data to deform one or more of the deformable mirrors of the deformable mirror array **126** such that the energy beam **104**, when generated or fired, will be directed to the predicted region. The preset configuration can minimize the movement required by the deformable mirror array **126** to achieve a final steering configuration that will direct the energy beam **104** to the strike location **148** once the strike location **148** is known based on the second tracking data.

[0067] At operation **804**, the second tracking system **146** generates the second tracking data representing the trajectory **142** of the target **106** inside of the reaction chamber **116**. The return light **149** can pass through the shutter apparatus **132** to arrive at the wavefront sensor array **122**, and the second tracking data can be generated based on the return light **149**. The second tracking data can represent the instantaneous position, velocity, or acceleration of the target **106** within the reaction chamber **116**. The second tracking data is generated based on the return light **149** traveling in line with the optical path **114**. Advantageously, monitoring the optical path **114** along which the energy beam **104** will eventually pass can ensure that localized variations in the atmosphere of the optical path **114** are inherently accounted for within the tracking data. More particularly, monitoring the

return light **149** that is in line with the energy beam **104**, rather than viewing the target **106** off axis from the optical path **114**, can provide beam steering based on the environment through which the energy beam **104** will travel, including the channel through the shutter apparatus **132**.

[0068] At operation **806**, the processing device determines, based on the second tracking data, the strike location **148** to be intercepted by the trajectory **142** of the target **106** within the strike zone **134**. The strike location **148** can be an instantaneous position of the target **106** within the strike zone **134** at which the energy beam **104** can be steered to intercept the target **106**.

[0069] At operation **808**, processing device actuates the beam steering system **124** of the fusion system **100** to steer the energy beam **104** along the optical path **114** to the strike location **148**. The deformable mirror array **126** can be adjusted to direct the energy beam **104** through the optical path **114** to intercept the target **106**. The energy beam **104** can pass through the shutter apparatus **132** when the shutter apparatus **132** is in a same open state during which return light **149** passed at operation **804**. The energy beam **104** can strike the target **106** within the reaction chamber **116** to initiate fusion.

[0070] As described above, the fusion system **100** can include precision target tracking, beam steering, and phase compensation capability. Real-time beam steering and phase compensation of each beamlet can be performed by the wavefront sensor array **122** and the deformable mirror array **126** operating in a high temporal bandwidth closed loop manner. Bulk subaperture spot positions on the wavefront sensor array **122** can queue global tilt correction offsets on the deformable mirror **126** while relative subaperture spot positions can queue spatially dependent actuator positions for higher order phase correction. A Shack-Hartmann wavefront sensor array **122** may be used having sufficient subapertures to balance signal to noise and having sufficient phase error detection for performing the methods described above. The wavefront sensor array **122** can be spectrally narrow band filtered centered on the target illumination and/or emission source and broadband neutral density filtered to prevent saturation and/or damage from other sources.

[0071] The method described above can be performed according to timing and synchronization considerations. Several such considerations are described below.

[0072] In an embodiment, the target pellet launch are nominally time synchronized with appropriate offsets for average time of flight to chamber center while accounting for the propagation time of laser pulses from the front-end pulse source locations to chamber center. More particularly, the speed of the target launch can be selected such that sufficient time exists to track the target and make the necessary corrections to cause the energy beam **104** to strike the target **106** at the strike location **148**.

[0073] In an embodiment, the energy pulse sources are time synchronized to all preamplification stages and final amplification stage SBS amplifier **130** (pump source) firing. The energy pulse source **102** can be initially queued from the first tracking data, e.g., generated by in-line cross cameras monitoring initial stage of target launch. For example, the first tracking data can be used to estimate a time of arrival of the target **106** at the strike zone **134** or the strike location **148** and the energy pulse source **102** can be activated or fired at a time to ensure that the energy beam **104** passes through the amplification stages to arrive at the strike location **148** simultaneously with the target **106**.

[0074] In an embodiment, the firing of a pump beam for the SBS amplification stage is synchronized to fire to optimize an overlap between seed beamlets and the pump beam. The overlap can occur at the final SBS amplification stage.

[0075] In an embodiment, the illumination sources, e.g., the glint illumination source, can be timed to ensure that return light **149** reflects from the target **106** when the target pellet is in flight. More particularly, the glint illumination can shine light that reflects as the return light **149** when the target **106** is visible within the optical path **114** through the shutter apparatus **132** and the entry port **136**.

[0076] In an embodiment, the shutter apparatus **132** is synchronized to ensure converging beamlets

pass forward when the shutter is in an open position. More particularly, the shutter ports **402**, **412** can remain in the open state at least until the energy beam **104** has passed through the channel into the lower pressure atmosphere. As described above, the shutter ports **402**, **412** can close after the energy beam **104** passes through the second shutter port **412** and before the gas from the higher pressure atmosphere has traversed the distance between the shutter ports **402**, **412** into the lower pressure atmosphere.

[0077] In an embodiment, the FLiBe nozzles (the nozzles used to open a port for the energy beam **104** and the return light **149** to pass into and out of the reaction chamber **116**) are synchronized to ensure that the target pellet and the laser engagement pulses are not blocked on respective entries into the reaction chamber. The nozzles can close the port immediately after passage of the energy beam **104** or the target **106** to then block entrance (or exit) of energy into or out of an interior of the chamber.

[0078] Referring to FIG. **9**, a block diagram of an example computing device that may perform one or more of the operations described herein is shown in accordance with an embodiment. A computing device **900** may be integrated in any of the fusion systems described above to perform any of the described operations. Computing device **900** may be connected to other computing devices in a LAN, an intranet, an extranet, and/or the Internet. The computing device may operate in the capacity of a server machine in the client-server network environment or in the capacity of a client in a peer-to-peer network environment. The computing device may be provided by a personal computer (PC), a set-top box (STB), a server, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single computing device is illustrated, the term “computing device” shall also be taken to include any collection of computing devices that individually or jointly execute a set (or multiple sets) of instructions to perform the methods discussed herein.

[0079] The example computing device **900** may include one or more processing devices **902** (e.g., a processing device, a general purpose processing device, a PLD, etc.), a main memory **904** (e.g., synchronous dynamic random access memory (DRAM), read-only memory (ROM)), a static memory **905** (e.g., flash memory and a data storage device), which may communicate with each other via a bus **930**.

[0080] The one or more processing devices **902** may be provided by one or more general-purpose processing devices such as a microprocessor, central processing unit, or the like. In an illustrative example, processing device(s) **902** may comprise a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processing device implementing other instruction sets or processing devices implementing a combination of instruction sets. Processing device(s) **902** may also comprise one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device(s) **902** may be configured to execute the method operations described herein, in accordance with one or more aspects of the present disclosure, for performing the operations and steps discussed herein.

[0081] Computing device **900** may further include a network interface device **908** which may communicate with a network **909**. The computing device **900** also may include a video display unit **910** (e.g., a liquid crystal display (LCD) or a cathode ray tube (CRT)), an alphanumeric input device **912** (e.g., a keyboard), a cursor control device **914** (e.g., a mouse) and an acoustic signal generation device **915** (e.g., a speaker). In one embodiment, video display unit **910**, alphanumeric input device **912**, and cursor control device **914** may be combined into a single component or device (e.g., an LCD touch screen).

[0082] Data storage device **918** may include a non-transitory computer-readable storage medium **928** on which may be stored one or more sets of instructions **925** that may include instructions for carrying out the operations described herein, in accordance with one or more aspects of the present

disclosure. Instructions **925** may also reside, completely or at least partially, within main memory **904** and/or within processing device(s) **902** during execution thereof by computing device **900**, main memory **904** and processing device(s) **902** also constituting computer-readable media. The instructions **925** may further be transmitted or received over a network **920** via network interface device **908**.

[0083] While computer-readable storage medium **928** is shown in an illustrative example to be a single medium, the term “computer-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database and/or associated caches and servers) that store the one or more sets of instructions. The term “computer-readable storage medium” shall also be taken to include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by the machine and that cause the machine to perform the methods described herein. The term “computer-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

[0084] The foregoing description, for the purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the embodiments and its practical applications, to thereby enable others skilled in the art to best utilize the embodiments and various modifications as may be suited to the particular use contemplated. Accordingly, the present embodiments are to be considered as illustrative and not restrictive, and the invention is not to be limited to the details given herein, but may be modified within the scope and equivalents of the appended claims.

Claims

1. A fusion system, comprising: a reaction chamber; an energy pulse source to generate an energy beam; a first tracking system to generate first tracking data based on a trajectory of a target outside of the reaction chamber; and a processing device, operatively coupled to the first tracking system and the energy pulse source, to: determine, based on the first tracking data, whether the trajectory is directed to a strike zone inside of the reaction chamber, and activate the energy pulse source in response to determining that the trajectory of the target is directed to the strike zone.
2. The fusion system of claim 1 further comprising: a second tracking system to generate second tracking data representing the trajectory of the target inside of the reaction chamber; and a beam steering system; wherein the processing device is further to: determine, based on the second tracking data, a strike location to be intersected by the trajectory of the target within the strike zone, and actuate the beam steering system to steer the energy beam along an optical path to the strike location.
3. The fusion system of claim 2, wherein the second tracking system includes a wavefront sensor array to generate the second tracking data based on light traveling proximally along the optical path from the target, and wherein the beam steering system includes a deformable mirror.
4. The fusion system of claim 2, wherein determining whether the trajectory is directed to the strike zone and controlling the beam steering system to steer the energy beam to the strike location occurs when the target is traveling from a target injection location outside of the reaction chamber to the strike location inside of the reaction chamber.
5. The fusion system of claim 2, wherein the processing device is to actuate the beam steering system to a preset configuration based on the first tracking data.
6. The fusion system of claim 2 further comprising a shutter apparatus along the optical path, wherein the shutter apparatus includes a shutter port, wherein return light travels proximally from the target through the shutter port, and wherein the energy beam travels distally through the shutter port.

7. The fusion system of claim 1 further comprising a second energy pulse source to generate a second energy beam toward the strike zone inside of the reaction chamber, wherein the processing device is to activate the second energy pulse source when the trajectory of the target is directed to the strike zone.

8. A target tracking system, comprising: a vacuum chamber; a first tracking system to generate first tracking data based on a trajectory of a target outside of the vacuum chamber; a laser pulse source to generate a laser beam; and a processing device, operatively coupled to the first tracking system and the laser pulse source, to: determine, based on the first tracking data, whether the trajectory is directed to a strike zone inside of the vacuum chamber, and activate the laser pulse source in response to determining that the trajectory of the target is directed to the strike zone.

9. The target tracking system of claim 8 further comprising: a second tracking system to generate second tracking data representing the trajectory of the target inside of the vacuum chamber; and a beam steering system; wherein the processing device is further to: determine, based on the second tracking data, a strike location to be intersected by the trajectory of the target within the strike zone, and actuate the beam steering system to steer the laser beam along an optical path to intersect the strike location.

10. The target tracking system of claim 9, wherein the second tracking system includes a wavefront sensor array to generate the second tracking data based on light traveling proximally along the optical path from the target, and wherein the beam steering system includes a deformable mirror.

11. The target tracking system of claim 9, wherein determining whether the trajectory is directed to the strike zone and controlling the beam steering system to steer the laser beam to the strike location occurs when the target is traveling from a target injection location outside of the vacuum chamber to the strike location inside of the vacuum chamber.

12. The target tracking system of claim 9, wherein the processing device is to actuate the beam steering system to a preset configuration based on the first tracking data.

13. The target tracking system of claim 9 further comprising a shutter apparatus along the optical path, wherein the shutter apparatus includes a shutter port, wherein return light travels proximally from the target through the shutter port, and wherein the laser beam travels distally through the shutter port.

14. The target tracking system of claim 8 further comprising a second energy pulse source to generate a second laser beam toward the strike zone inside of the vacuum chamber, wherein the processing device is to activate the second energy pulse source when the trajectory of the target is directed to the strike zone, and to not activate the second energy pulse source when the trajectory of the target is not directed to the strike zone.

15. A method, comprising: generating, by a first tracking system of a fusion system, first tracking data based on a trajectory of a target outside of a reaction chamber of the fusion system; determining, by a processing device of the fusion system based on the first tracking data, whether the trajectory is directed to a strike zone inside of the reaction chamber; and activating an energy pulse source of the fusion system to generate an energy beam in response to determining that the trajectory of the target is directed to the strike zone.

16. The method of claim 15 further comprising: generating, by a second tracking system of the fusion system, second tracking data representing the trajectory of the target inside of the reaction chamber; determining, by the processing device based on the second tracking data, a strike location to be intersected by the trajectory of the target within the strike zone; and actuating, by the processing device, a beam steering system of the fusion system to steer the energy beam along an optical path to the strike location.

17. The method of claim 16, wherein the second tracking system includes a wavefront sensor array to generate the second tracking data based on light traveling proximally along the optical path from the target, and wherein the beam steering system includes a deformable mirror.

18. The method of claim 16, wherein determining whether the trajectory is directed to the strike

zone and controlling the beam steering system to steer the energy beam to the strike location occurs when the target is traveling from a target injection location outside of the reaction chamber to the strike location inside of the reaction chamber.

19. The method of claim 16 further comprising actuating, by the processing device, the beam steering system to a preset configuration based on the first tracking data.

20. The method of claim 16 further comprising opening, by the processing device, a shutter apparatus along the optical path, wherein the shutter apparatus includes a shutter port, wherein return light travels proximally from the target through the shutter port, and wherein the energy beam travels distally through the shutter port.
